

Survey / Review Report Assignment

Topic Selection

Choose **one** of the following topics for your survey/review report:

- Deep Learning-based Medical Image Segmentation
- Deep Learning-based Tumor Detection
- Multimodal Large Language Models (LLMs) in Medical Imaging

Note: For your selected topic, you may either conduct a general survey or focus on a specific organ and imaging modality, such as brain MRI, lung CT, or breast mammography.

Report Requirements

- **Length:**
The main body of the report should be approximately **6 pages**, excluding the cover page and references.
- **Template:**
Use the official LaTeX template provided by the course:
[IMC LaTeX Thesis Template](#)
- **Academic Integrity:**
The report must be written independently by the student(s). The use of AI-generated content (e.g., ChatGPT or other generative AI tools) to write the report is not permitted. Any external sources used for ideas, figures, or references must be properly cited.

Recommended Report Structure

Your report should include the following sections:

1. Introduction / Background

Provide a clear introduction to the selected topic.

2. State of the Art (SOTA)

Review and summarize important existing methods, models, and research papers related to the topic. Compare approaches where appropriate.

3. Challenges

Discuss major technical and practical challenges in the field, such as:

- Data scarcity
- Annotation cost
- Model generalization

- Interpretability
- Clinical deployment limitations

4. Conclusion and Discussion

Summarize the main findings from the literature and critically discuss:

- Strengths of current approaches
- Limitations
- Clinical and research implications

5. Future Ideas / Outlook

Present potential future research directions, improvements, or extensions related to the topic.

6. Self-Reflection Note

Include a **half-page reflection** describing:

- Your learning process
- Key insights gained

7. References and Citations

- Select references carefully, prioritizing high-quality, recent, and relevant academic sources.
- Use **proper academic citations** throughout the report.
- All figures, tables, and external resources must be referenced correctly.
- Follow a consistent citation style (e.g., IEEE, APA, or BibTeX format in LaTeX).